Energy Consumption & Analysis Query

Check the historical data and see how I can make the best use of the energy generated from my PV panel.

Intent Classification

```
{ "Primary": "Energy Consumption and Analysis",
  "Secondary": "Energy Optimization" }
```

Code Interpreter

```
file_path = '/mnt/data/file-name'
data = pd.read_csv(file_path)
data.head()
...
```

Based on the data, I will conduct the following steps:
1. Aggregate and Analyze Local and Grid Energy Data
2. Visualize Energy Usage Patterns
3. Provide Optimization Suggestions

…

Code Interpreter

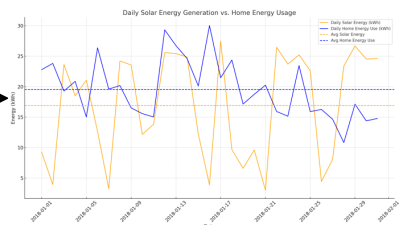
```
total_local_kWh = local_data['total_local_kWh'].sum()
total_grid_kWh = grid_data['total_grid_kWh'].sum()
...
```

Summary of energy data:…
Key observations: …
Next steps: 1. Visualize Daily Energy Generation and Total Usage. 2. Optimize Home Energy Usage

…

Code Interpreter

```
plt.figure(figsize=(14, 8))
plt.plot(…)
plt.savefig('/mnt/data/daily
_local_vs_home_energy.png')
plt.show()
```



The graph shows a comparison between the daily energy generation and the total home energy usage.
Recommendations to Optimize Energy Usage (shortened):
1. Load Shift Major Appliances
2. Battery Storage Integration

…

Device Status & Control Query

Set the living room light to a brightness level good for reading.

Intent Classification

```
{ "Primary": "Device Status and Control",
  "Secondary": "Device Custom Configurations" }
```

Let's conduct the following steps:
1. Synchronize device information
2. Query the current state of the Living Room Light
3. Execute the command to set appropriate brightness

Action Device Sync

```
{"device_name":"Livingroom Light"}
```

Smart Device API
Simulated by JSON Files

```
{'name': 'Livingroom light',
 'type':
'action.devices.types.LIGHT',
 'traits': ['ColorSetting',
'Brightness', 'OnOff']…
```

Action Device Query

```
{"device_name":"Livingroom Light"}
```

Smart Device API
Simulated by JSON Files

```
{'name': 'Livingroom light',
 'type': 'action.devices.types.LIGHT',
 'status': 'SUCCESS',
 'online': True,
 'on': True,
 'brightness': 50, …
```

Action Device Execute

```
{"device_name":"Livingroom Light",
 "parameter":"brightness",
 "update_value": 75}
```

Smart Device API
Simulated by JSON Files

```
{'update': 'SUCCESS'}
```

The **living room light** is now set to a **brightness level of 75%,** which is suitable for reading. If you need any further adjustments, feel free to let me know!